

Stochastic processes

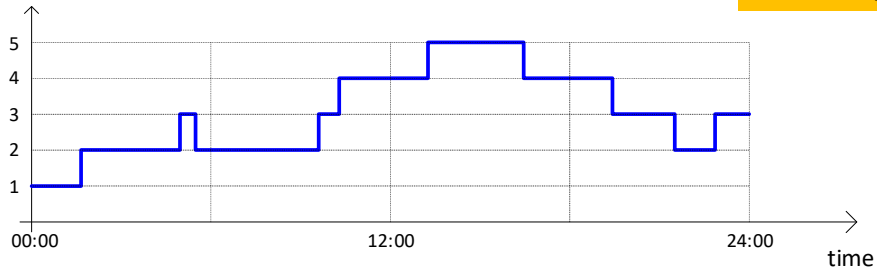
Stochastic processes

- Stochastic processes model **randomness along time**, or another 2nd dimension
- Examples:
 - Number of phone calls served by a network operator, along time
 - Number of clients waiting in a queuing system, along time
 - Duration of the successive calls served by a network operator, indexed according to the order of arrival (2nd dimension is not time)
 - Delay suffered by successive clients in a queuing system, indexed by arrival order (2nd dimension is not time)
 - Weather conditions (raining or not raining) along successive days
 - Number of attacks to a server along successive days
- The observations of stochastic processes are time functions (or functions of a 2nd dimension) – **sample paths**

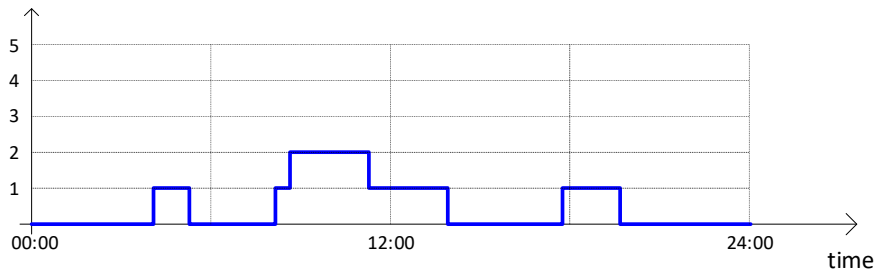
Stochastic processes

Number of calls served by a network operator along time
– 3 sample paths

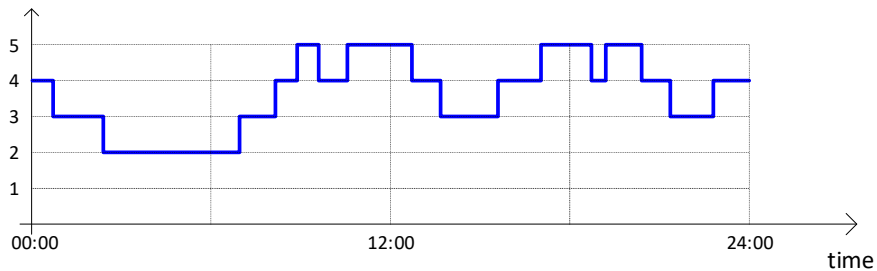
Day 1



Day 2



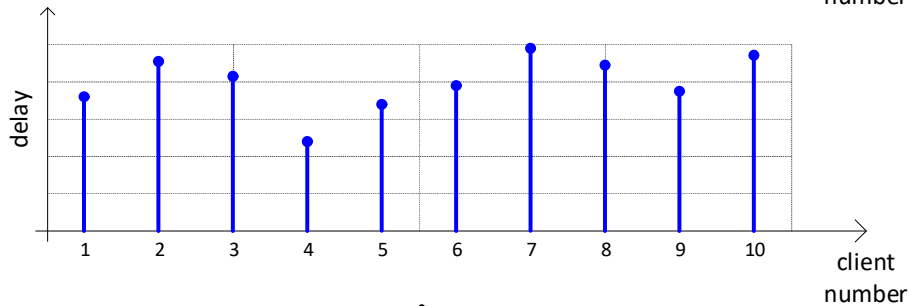
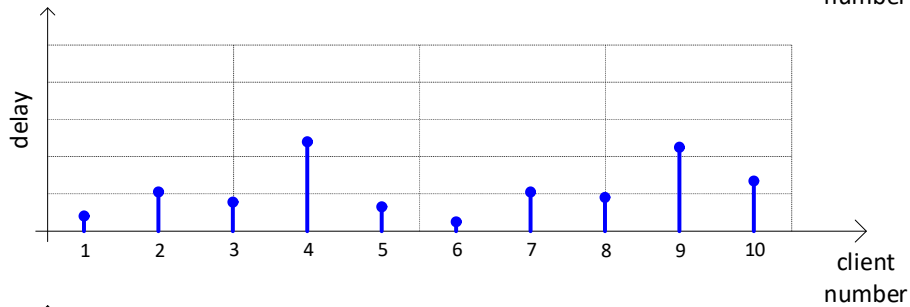
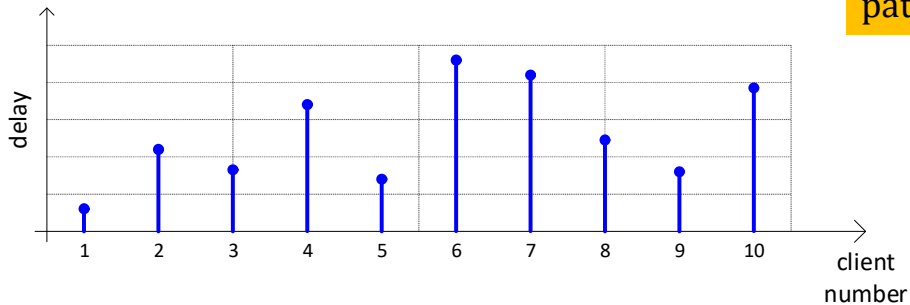
Day 3



⋮

Stochastic processes

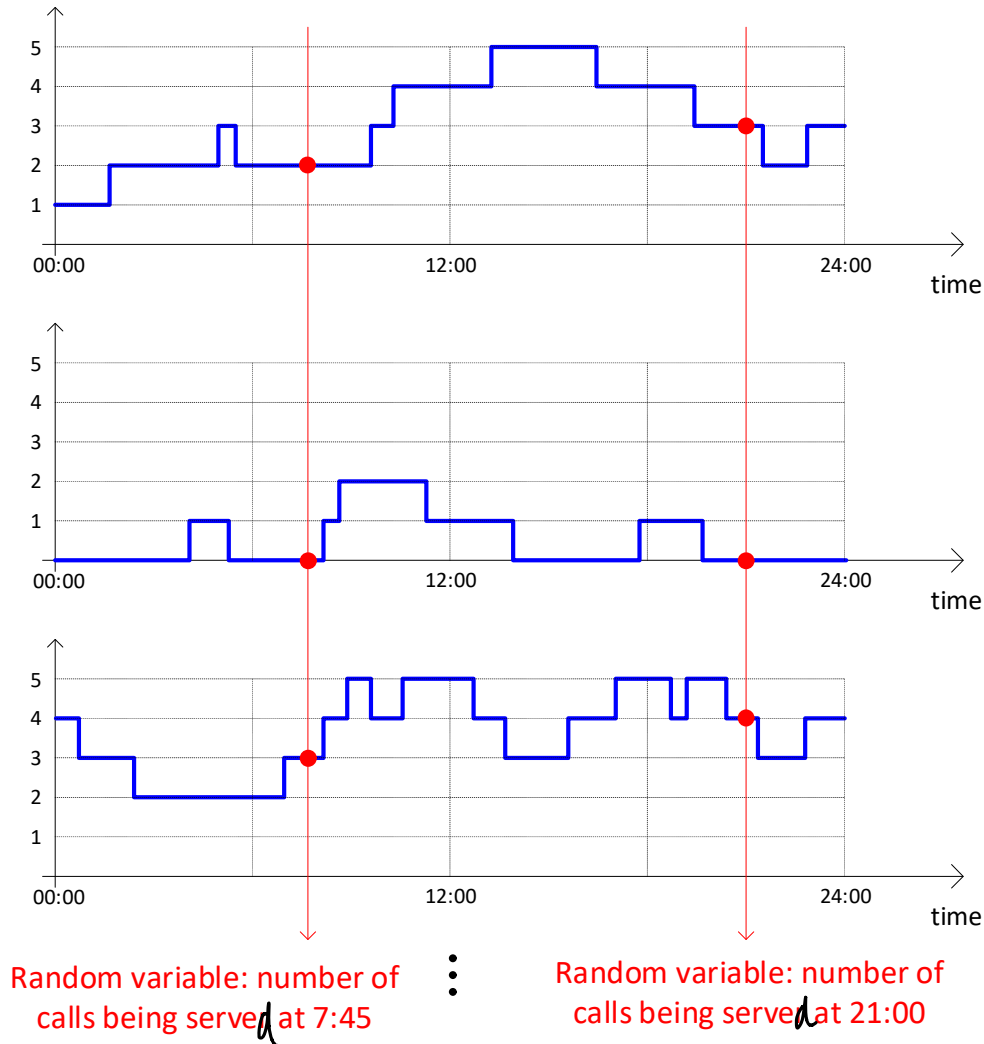
Delays of successive clients in
queuing system – 3 sample
paths



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Stochastic processes

For each $t \in T$, $X(t)$ is a random variable

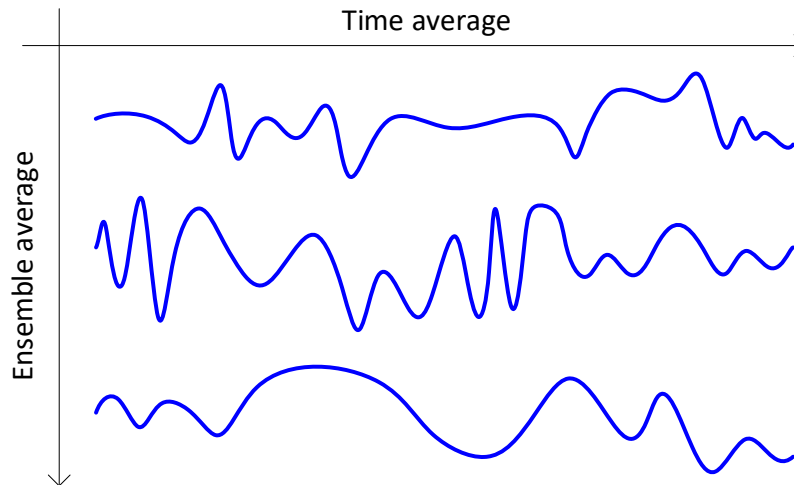


Stochastic processes

- A **stochastic process** $\{X(t), t \in T\}$ is a set of random variables: for each $t \in T$, $X(t)$ is a random variable.
- The index t is frequently interpreted as time. In this case, $X(t)$ is the state of the process at time t .
- T is the index set of the process
 - If T is a countable set, the process is said in **discrete time**
 - If T is an interval of the real axis, the process is said in **continuous time**
- The **state space** is the set of values that the random variables $X(t)$ can take.

Stochastic processes

- **Ensemble average:** average over the ensemble of sample paths; may vary with t
- **Time average:** average over time of a specific sample path



Stochastic processes

- A stochastic process is said **stationary** if the distribution of its random variables does not vary with time
- A stochastic process is said **ergodic** if its statistical properties can be deduced from a single, sufficiently long, sample path
 - A stochastic process is said **mean-ergodic** if the time average of any sample path tends to the ensemble average, as $N \rightarrow \infty$.